23.	(a) Find $L\left[\frac{\cos at - \cos bt}{t}\right]$	8	3	3
	(OR)			
	(b) Find $L^{-1}\left[\log\left(\frac{s^2+1}{s^2-1}\right)\right]$			
24.	(a) If f(z) is analytic function of z, prove that $\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)\lvert f(z)\rvert^2 = 4\,\lvert f'(z)\rvert^2$	8	3	4
	(OR)			
	(b) Determine the analytic function $f(z) = u + iv$ such that $u - v = e^x(\cos y - \sin y)$			
25.	(a) Expand the function $\frac{1}{z^2-3z+2}$ by Laurent's series in the regions (i) $1 < \lvert z\rvert < 2$ (ii) $1 < \lvert z-1\rvert < 2$	8	3	5
	(OR)			
	(b) Evaluate $\int_c \frac{\cos \pi z^2}{(z-1)(z-2)}dz$ where C is $\lvert z\rvert = \frac{3}{2}$.			

PART - C (1 × 15 = 15 Marks)

Answer **any 1** Questions

		Marks	BL	CO
26.	Using Laplace transform, Solve the differential equation $\frac{d^2y}{dt^2} - 4\frac{dy}{dt} + 3y = e^{-t}$, given $y(0) = 1$ and $y'(0) = 0$.	15	3	3
27.	Evaluate $\int_0^{2\pi} \frac{d\theta}{13 + 5\sin\theta}$, by using contour integration.	15	5	5

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Reg. No

B.Tech/M.Tech (Integrated) DEGREE EXMINATION, JULY 2025

Second Semester

21MAB102T - ADVANCED CALCULUS AND COMPLEX ANALYSIS

(For the candidates admitted during the academic year 2021-2022 to 2024-2025)

Note:

i. **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.

ii. **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 hours **Max. Marks: 75**

PART - A (20 × 1 = 20 Marks)

Answer **all** Questions

		Marks	BL	CO
1.	The value of $\int_1^a \int_1^b \frac{dx\,dy}{xy}$ is (A) loga logb (B) a.b (C) log(ab) (D) log a+b	1	1	1
2.	Changing to polar coordinates, the double integral $\int_0^a \int_y^a f(x,y)\,dx\,dy$ becomes (A) $\int_0^{\frac{\pi}{4}} \int_0^{a\sin\theta} \phi(r,\theta)\,dr\,d\theta$ (B) $\int_0^{\frac{\pi}{4}} \int_0^{a\sin\theta} \phi(r,\theta) r\,dr\,d\theta$ (C) $\int_0^{\frac{\pi}{4}} \int_0^{a\sec\theta} \phi(r,\theta) r\,dr\,d\theta$ (D) $\int_0^{\frac{\pi}{4}} \int_0^{a\sec\theta} \phi(r,\theta)\,dr\,d\theta$	1	1	1
3.	The integral $\int_0^1 \int_x^1 dy\,dx$, after the change of order of integration becomes (A) $\int_0^x \int_0^y dx\,dy$ (B) $\int_0^1 \int_0^1 dx\,dy$ (C) $\int_0^x \int_0^1 dx\,dy$ (D) $\int_0^1 \int_0^y dx\,dy$	1	1	1
4.	$\int_0^1 \int_0^1 \int_0^1 e^{x+y+z} dy\,dx\,dz$ is equal to (A) $(e-1)^3$ (B) e^3 (C) $(e-1)^2$ (D) $3e$	1	1	1
5.	If $\phi = x^2 + y^2 + z^2 - 8$, then grad ϕ at (2, 0, 2) (A) $\vec{i} + 4\vec{k}$ (B) $4\vec{i} + 4\vec{k}$ (C) $4\vec{i} + \vec{k}$ (D) $4\vec{i} + 4\vec{j} + 4\vec{k}$	1	1	2

6. If $\phi = x^2 - y^2$, then $\nabla^2\phi = \dfrac{\partial^2\phi}{\partial x^2} + \dfrac{\partial^2\phi}{\partial y^2}$ is equal to 1 1 2

(A) 0 (B) 2
(C) -2 (D) 1

7. The unit normal vector to the surface $x^2 + y^2 = z$ at the point (1, 2, 5) is 1 1 2

(A) $\dfrac{2\vec{i} + 4\vec{j} - \vec{k}}{\sqrt{21}}$ (B) $\dfrac{1}{\sqrt{3}}\vec{i} + \dfrac{1}{\sqrt{3}}\vec{j} - \dfrac{1}{\sqrt{3}}\vec{k}$
(C) $\dfrac{\vec{i} + 4\vec{j} - 5\vec{k}}{\sqrt{42}}$ (D) $\dfrac{1}{\sqrt{2}}\vec{i} - \dfrac{1}{\sqrt{2}}\vec{k}$

8. If $\vec{F} = x^2\vec{i} + xy^2\vec{j}$, then value of the integral $\int_c \vec{F}.\vec{dr}$, where C is the segment on y = x from (0, 0) to (1, 1) is 1 1 2

(A) $-\dfrac{7}{6}$ (B) $\dfrac{7}{12}$
(C) $\dfrac{7}{6}$ (D) $-\dfrac{7}{12}$

9. $L[t^2 2^t]$ is 1 1 3

(A) $\dfrac{1}{(s-2)^2}$ (B) $\dfrac{1}{(s-log2)^2}$
(C) $\dfrac{2}{(s-log\,2)^3}$ (D) $\dfrac{2}{(s-2)^2}$

10. $L[e^{2t}\sin t]$ is 1 1 3

(A) $\dfrac{s}{(s-2)^2}$ (B) $\dfrac{1}{(s-2)^2+1}$
(C) $\dfrac{s}{(s-2)^2+1}$ (D) $\dfrac{1}{(s-2)^2}$

11. If $L[f(t)] = \dfrac{s}{(s+2)^2}$, then $f(t)$ is equal to 1 1 3

(A) $e^t\cos t$ (B) $e^{-2t}.t$
(C) $e^{-2t}(1-2t)$ (D) $e^{-t}2t$

12. $L\left[\int_0^t e^{-t}\cos t\,dt\right]$ is 1 1 3

(A) $\dfrac{s+1}{s^2+2s+2}$ (B) $\dfrac{s+1}{s(s^2+2s+2)}$
(C) $\dfrac{s+1}{s(s+1)^2}$ (D) $\dfrac{s+1}{(s+1)^2}$

13. If $f(z) = u + ic$ is analytic, where c is a constant, then u is 1 1 4

(A) analytic (B) Constant
(C) differentiable (D) harmonic

14. The function $f(z) = |z|^2$ 1 1 4

(A) constant (B) is analytic
(C) does not satisfy the C-R equations at (0, 0) (D) is not analytic

15. The invariant points of the transformation $w = \dfrac{3z-5}{z+1}$ are 1 1 4

(A) $i, -i$ (B) $2i, -2i$
(C) $1+2i, 1-2i$ (D) $2-i, 2+i$

16. The image of the circle $|z| = 2$ under the map $w = \dfrac{1}{z}$ is 1 1 4

(A) a circle of radius 2 (B) a circle of radius $\dfrac{1}{2}$
(C) a circle through the origin (D) unit circle

17. The order of the pole z = 0 for $\dfrac{1-e^z}{z^4}$ is 1 1 5

(A) 2 (B) 3
(C) 5 (D) 4

18. For $\dfrac{\sin z}{z}, z = 0$ is ________ singularity. 1 1 5

(A) removable singularity (B) pole
(C) essential singularity (D) Not a singularity

19. The Poles of $f(z) = \dfrac{z^2-1}{(z-1)^2(z+2)}$ are 1 1 5

(A) z = 1, −2 are of order 2 (B) z = 1 is of order 2, z = −2 is of order 1
(C) z = 1, −2 are of order 1 (D) z = 1 is of order 1, z = −2 is of order 2

20. The value of the integral $\int_c \dfrac{\cos\pi z}{z-1}dz$, if C is $|z| = \dfrac{3}{2}$, is 1 1 5

(A) $2\pi i$ (B) $4\pi i$
(C) πi (D) $-2\pi i$

PART - B (5 × 8 = 40 Marks)
Answer **all** Questions

	Marks	BL	CO

21. (a) Change the order of integration in the following integral and hence evaluate $\int_0^a\int_y^a \dfrac{x}{x^2+y^2}dx\,dy.$ 8 3 1

(OR)

(b) Evaluate $\int_0^4\int_0^{2\sqrt{z}}\int_0^{\sqrt{4z-x^2}} dy\,dx\,dz$

22. (a) Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $x^2 + y^2 - z = 3$ at the point (2, -1, 2) 8 3 2

(OR)

(b) Evaluate Green's theorem in plane $\iint (x^2 - y^2)dx + 2xydy$ where C is the boundary of the rectangle in the XoY plane bounded by the lines x=0, x=a, y=0, y=b.